

P8131 Spring 2026 Homework #1

Due on February 9 11:59pm

1. Show that the following distributions belong to the exponential family. Find the natural parameter θ , scale parameter ϕ and convex function $b(\theta)$. Also find the EY and $\text{Var}(Y)$ as functions of the natural parameter. Specify the canonical link functions.
 - (a) Exponential distribution $Exp(\lambda)$, $f(y; \lambda) = \lambda e^{-\lambda y}$;
 - (b) Binomial distribution $Bin(n, \pi)$, $f(y; \pi) = \binom{n}{y} \pi^y (1 - \pi)^{n-y}$, where n is known;
 - (c) Poisson distribution $Pois(\lambda)$, $f(y; \lambda) = \frac{1}{y!} \lambda^y e^{-\lambda}$;
 - (d) Negative binomial distribution $NB(m, \beta)$, $f(y; \beta) = \binom{y+m-1}{m-1} \beta^m (1 - \beta)^y$, where m is known;
 - (e) The Gamma distribution $Gamma(\alpha, \beta)$, $f(y; \beta) = \frac{\beta^\alpha}{\Gamma(\alpha)} y^{\alpha-1} e^{-\beta y}$, where the shape parameter α is known.
2. Assume $Y_1, Y_2, \dots, Y_n$ are independent and follow a binomial distribution where $Y_i \sim Bin(m, \pi_i)$ and m is known. Furthermore, assume $\log \frac{\pi_i}{1-\pi_i} = X_i \beta$. What are the expressions of deviance residuals and Pearson residuals respectively (use $\hat{\beta}$ to represent the MLE)? What are the expressions of the deviance and Pearson's χ^2 statistic?
3. Consider the binary response variable $Y \sim Bernoulli$ with $P(Y = 1) = \pi$ and $P(Y = 0) = 1 - \pi$. Observations $Y_i, i = 1, \dots, n$, are independent and identically distributed as Y .
 - (a) Find the Wald test statistic, the score test statistic, and the likelihood ratio test statistic to test hypothesis $H_0 : \pi = \pi_0$.
 - (b) With large samples, the Wald test statistic, score test statistic and the likelihood ratio test statistic approximately have the $\chi^2(1)$ distribution. For $n = 10$ and data $(0, 1, 0, 0, 1, 0, 0, 0, 1, 0)$, use these statistics to test null hypotheses on for (i) $\pi_0 = 0.1$, (ii) $\pi_0 = 0.3$, (iii) $\pi_0 = 0.5$.
 - (c) Do the Wald test, score test, and the likelihood ratio test lead to the same conclusions in (b)?
4. Assume $Y_i \sim Pois(\lambda), i = 1, \dots, n$. We are interested in testing $H_0 : \log \lambda = \log \lambda_0$. What are the Wald test statistic, the score test statistic, and the likelihood ratio test statistic?